
Search Problems

CS5491: Artificial Intelligence
JIANYUAN GUO, ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



TODAY

Agents that Plan Ahead

Search Problems

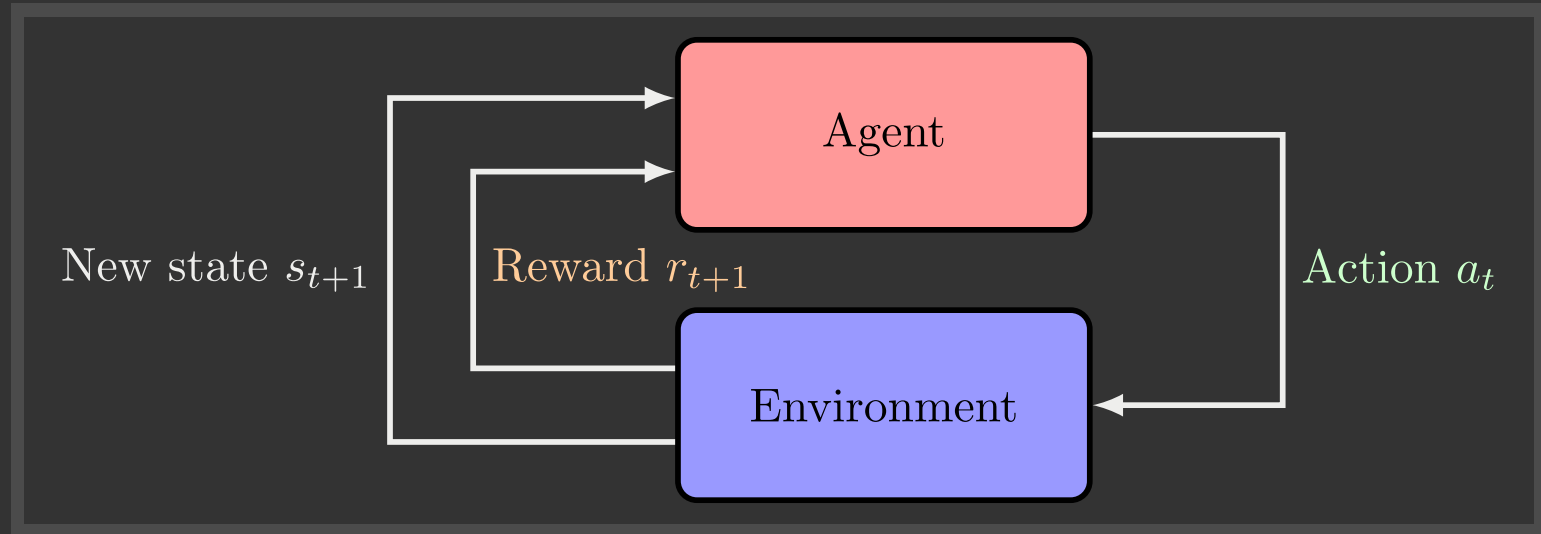
Reading

- Today's Lecture: RN Chapter 3.1-3.4
- Next Lecture: RN Chapter 3.1-3.4

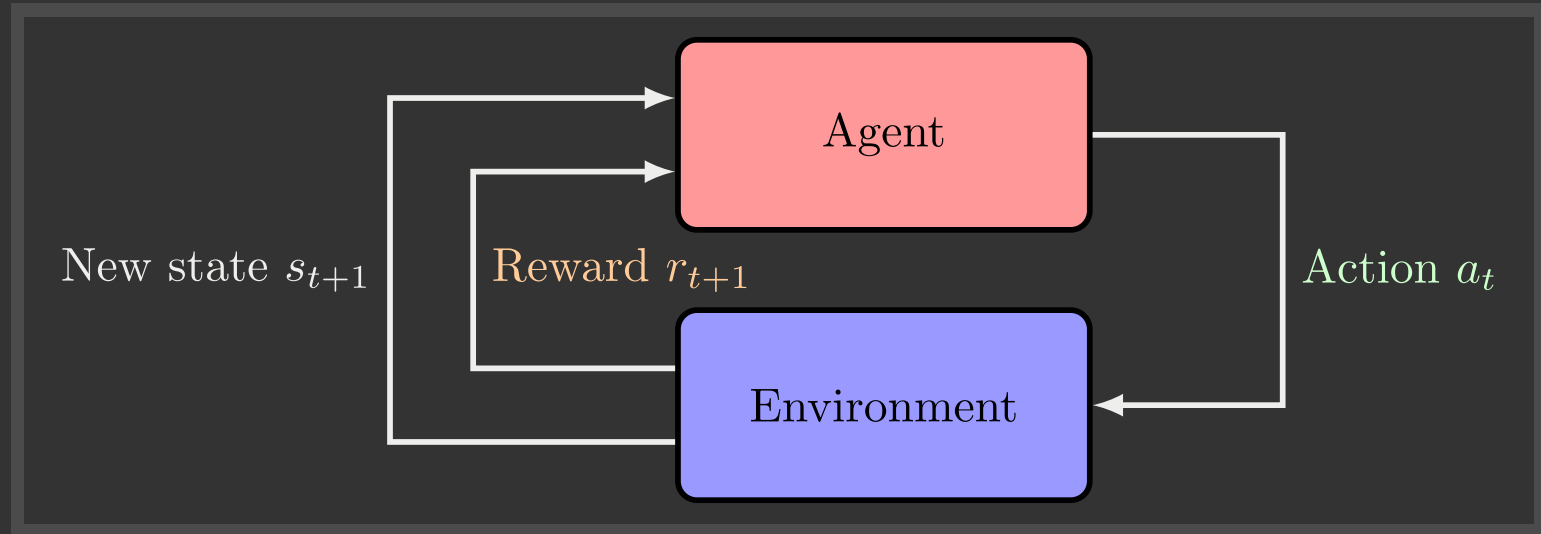


XKCD

REMINDER: RATIONAL AGENTS

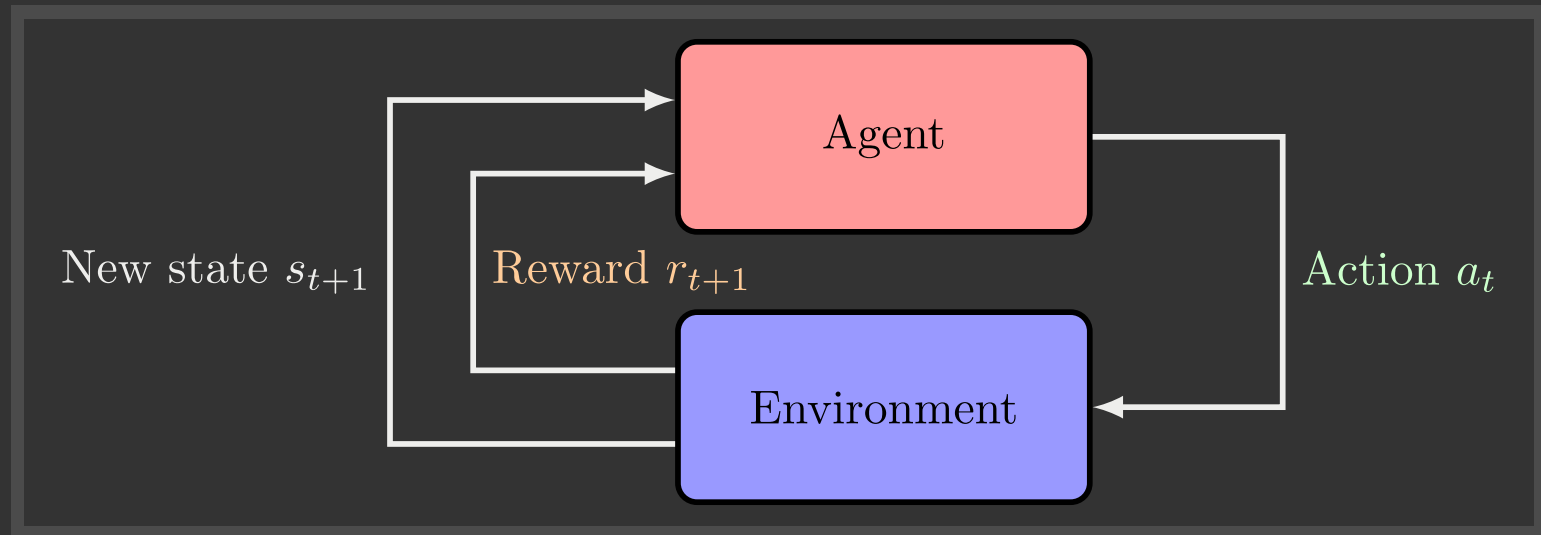


REMINDER: RATIONAL AGENTS



An **agent** is an entity that *perceives* and *acts*.

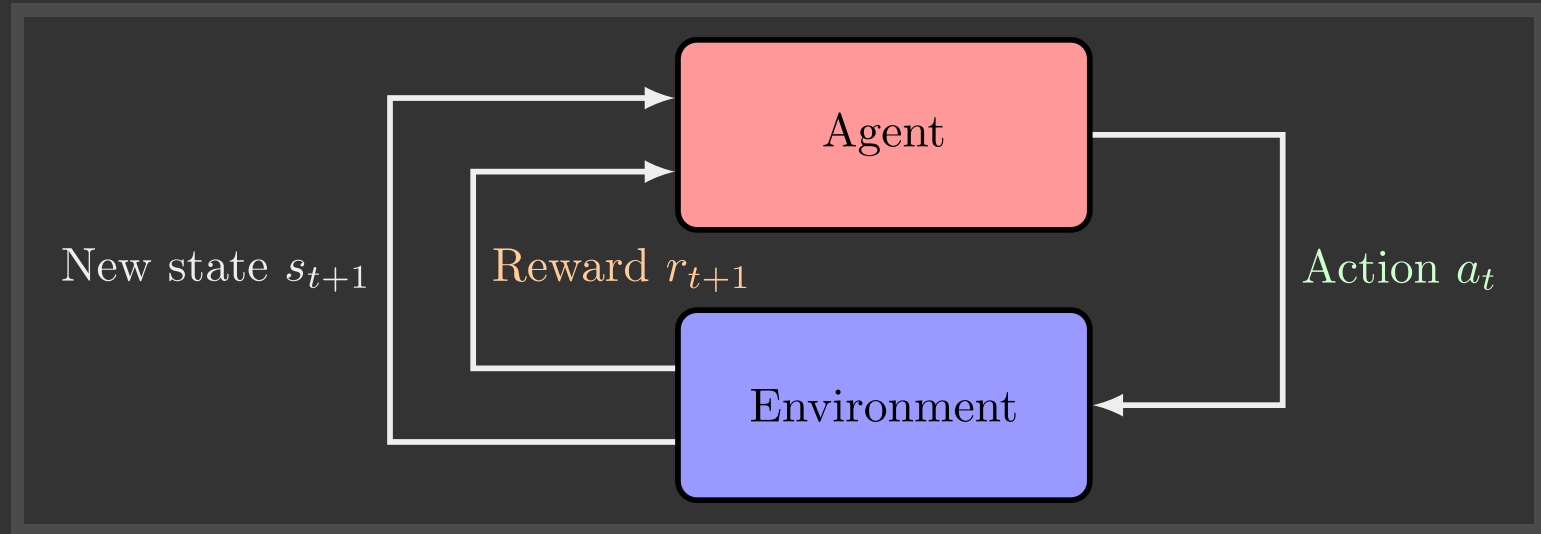
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A **rational agent** selects actions that maximize its (expected) **utility**.

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A **rational agent** selects actions that maximize its (expected) **utility**.

Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions.

RATIONAL AGENTS



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Do rational agents **explore** and **learn**?

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So, rational agents are not necessarily successful, but they are **autonomous**.



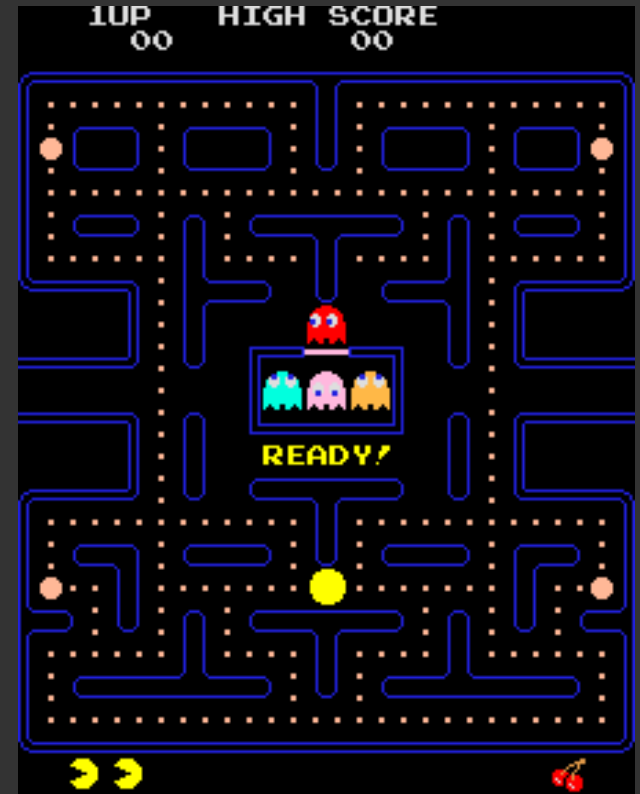
RATIONAL AGENTS

Performance Measure

Environment

Actuators

Sensors



RATIONAL AGENTS

Performance Measure

- -1 per time step; +10 food; +500 win; -500 die; +200 hit scared ghost

Environment

- Pacman dynamics + ghost behavior

Actuators

- North, West, East, South, Stop (idle)

Sensors

- entire state is visible



ROBOTAXI - PEAS

Performance Measure

Environment

Actuators

Sensors



ROBOTAXI - PEAS

Performance Measure

- income, happy customer, vehicle costs, fines, car, insurance

Environment

- streets, other drivers, customers, etc.

Actuators

- steering, brake, gas, display/speaker

Sensors

- cameras, lidar, radar, ultrasonic, accelerometer, mechanical sensors, microphone, etc.



ENVIRONMENT CATEGORIZATION

	Pacman	Robotaxi
Fully or Partially Observable		
Single or Multi Agent		
Deterministic or Stochastic		
Static or Dynamic		
Discrete or Continuous		



ENVIRONMENT CATEGORIZATION

	Pacman	Robotaxi
Fully or Partially Observable	fully	partial
Single or Multi Agent	multi	multi
Deterministic or Stochastic	deterministic	stochastic
Static or Dynamic	static	dynamic
Discrete or Continuous	discrete	continuous



ENVIRONMENT CATEGORIZATION

Reflex agents:

- Choose actions based on current observation (and maybe memory)
- May have memory or a model of the world's current state.
- Do not consider future consequences of actions
- Consider how the world IS as opposed to how it would be.

Can a reflex agent be rational?



AGENTS THAT PLAN AHEAD

Planning Agents:



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Planning Agents:

- Decision based on predicted consequences of actions



AGENTS THAT PLAN AHEAD

Planning Agents:

- Decision based on **predicted consequences** of actions
- Must have a **transition model** : how the world evolves in responses to actions



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Spectrum of Deliberativeness:



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Spectrum of Deliberativeness:

- Generate complete, optimal plan offline, then execute



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Planning Agents:

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- Must formulate a goal
- **Consider how the world WOULD BE as opposed to how it is**

Spectrum of Deliberativeness:

- Generate complete, optimal plan offline, then execute
- Generate simple, greedy plan, start executing, replan if necessary



SEARCH PROBLEMS



WHAT ARE SEARCH PROBLEMS?

A **search problem** consists of:



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A **search problem** consists of:

- A state space:



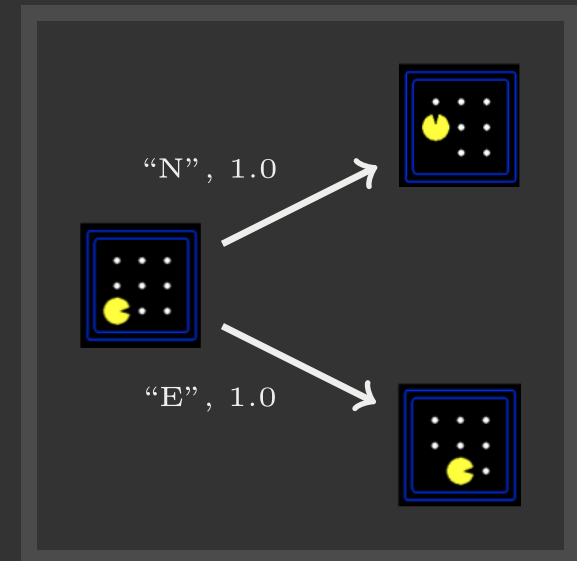
WHAT ARE SEARCH PROBLEMS?

A **search problem** consists of:

- A state space:



- A successor function (actions, costs):



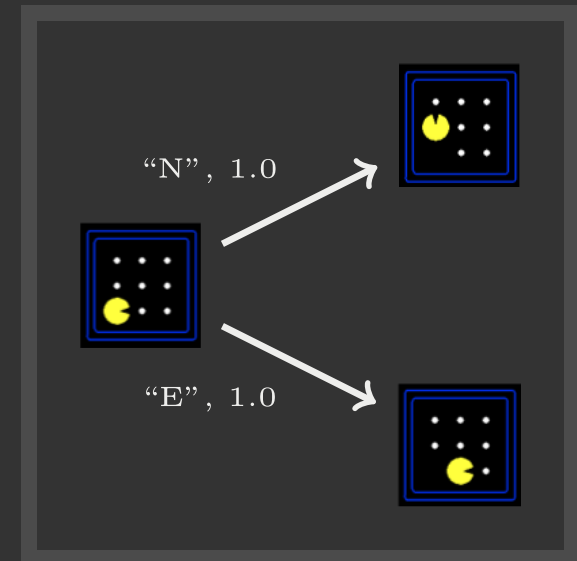
WHAT ARE SEARCH PROBLEMS?

A **search problem** consists of:

- A state space:



- A successor function (actions, costs):
- A start state and goal test



WHAT ARE SEARCH PROBLEMS?

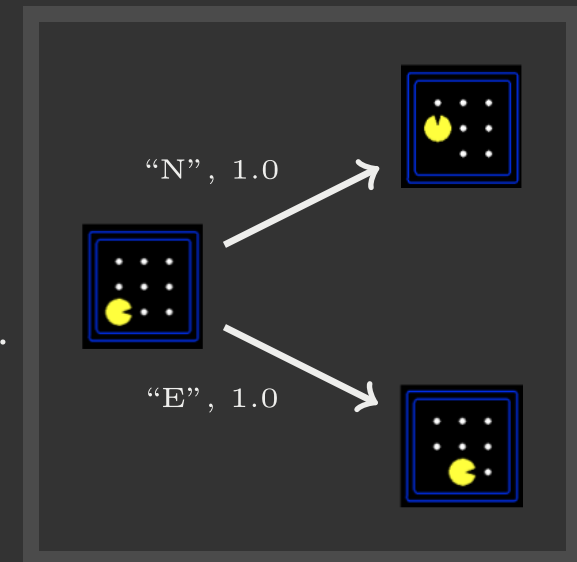
A **search problem** consists of:

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- A start state and goal test

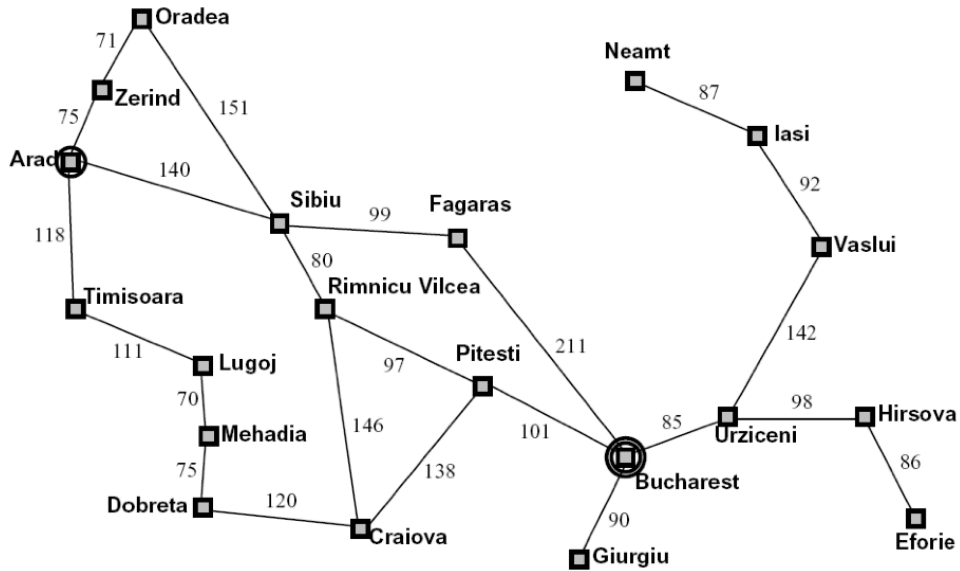
A **solution** is a sequence of actions (a plan) which transforms the start state to the goal test.



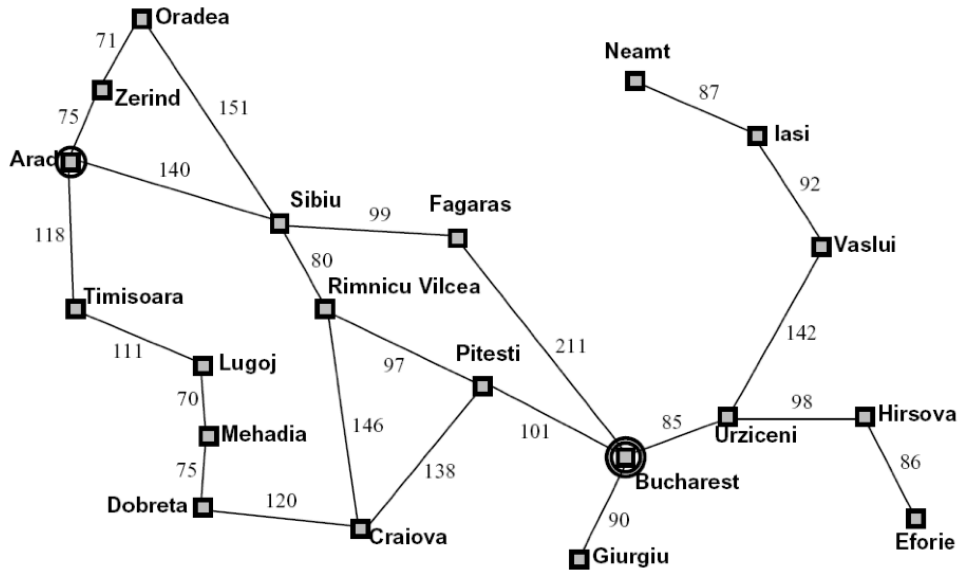
SEARCH PROBLEMS ARE MODELS



EXAMPLE: TRAVELING IN ROMANIA



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State Space:

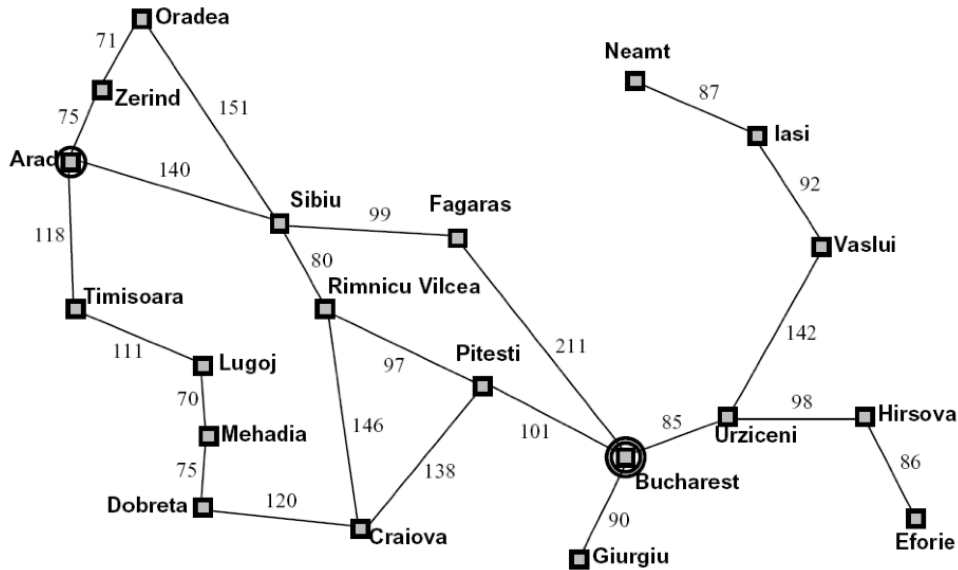
Successor Function:

Start Space:

Goal Test:

Solution?

EXAMPLE: TRAVELING IN ROMANIA



State Space:

- Cities

Successor Function:

- Roads: travel to adjacent city with cost=distance

Start Space:

- Arad

Goal Test:

- Is state == Bucharest?

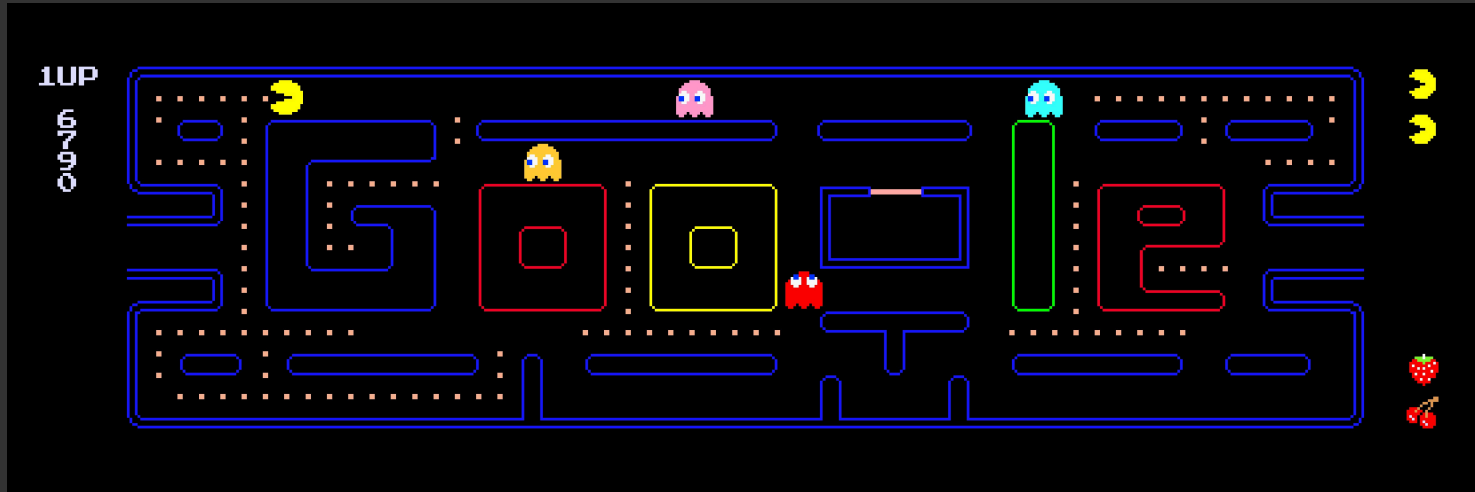
Solution?

WHAT IS IN A STATE SPACE?



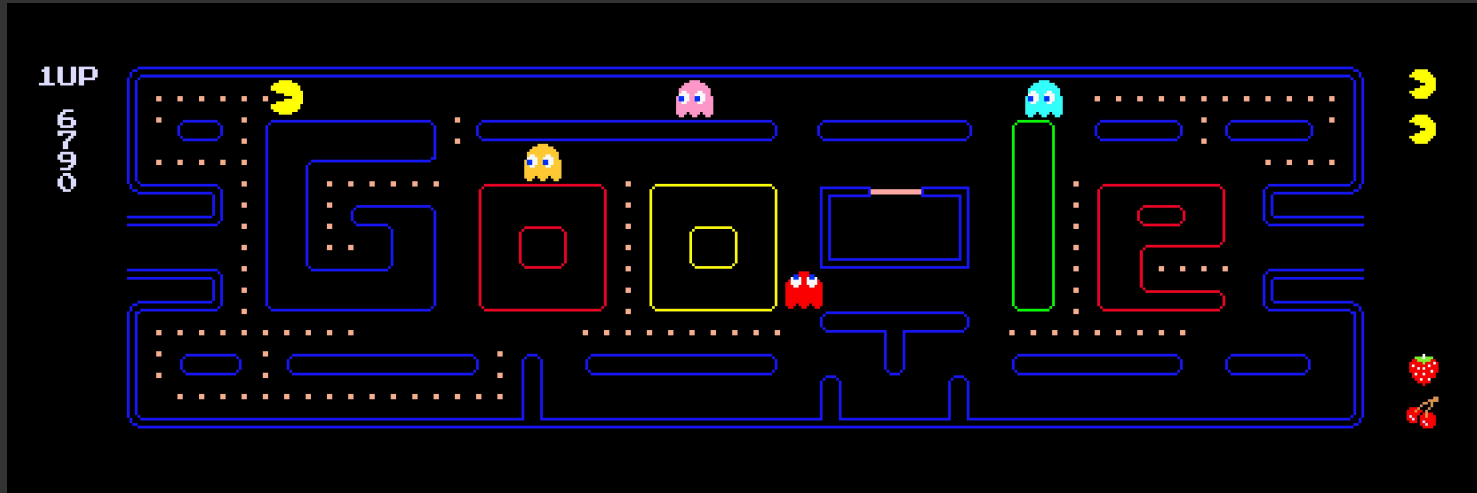
WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



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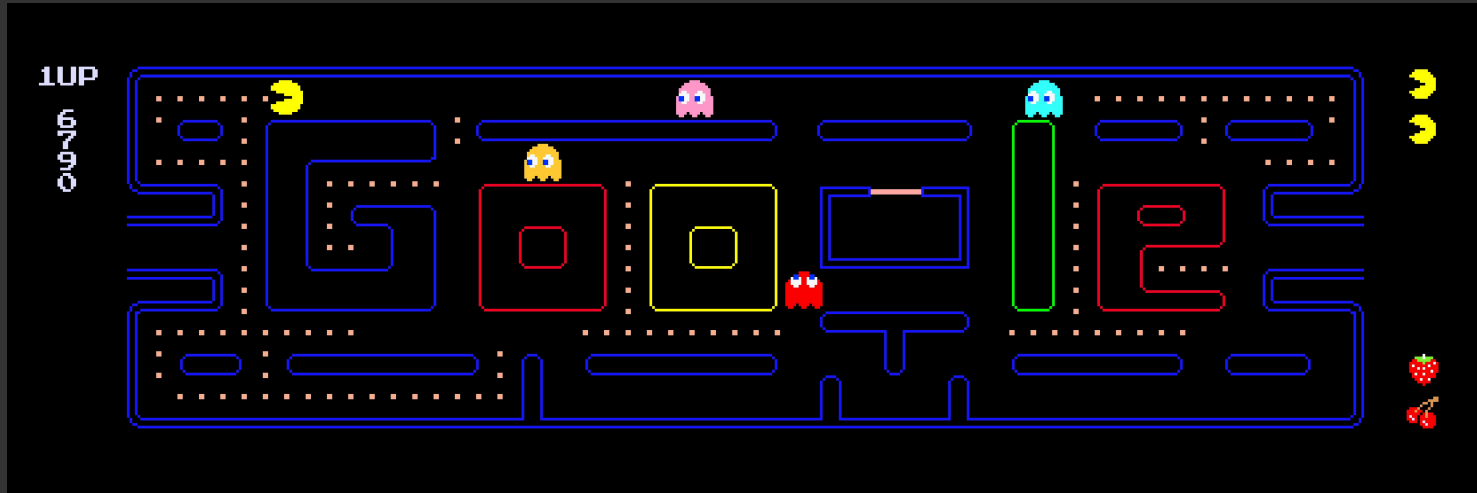
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Search State: Keeps only details necessary for planning

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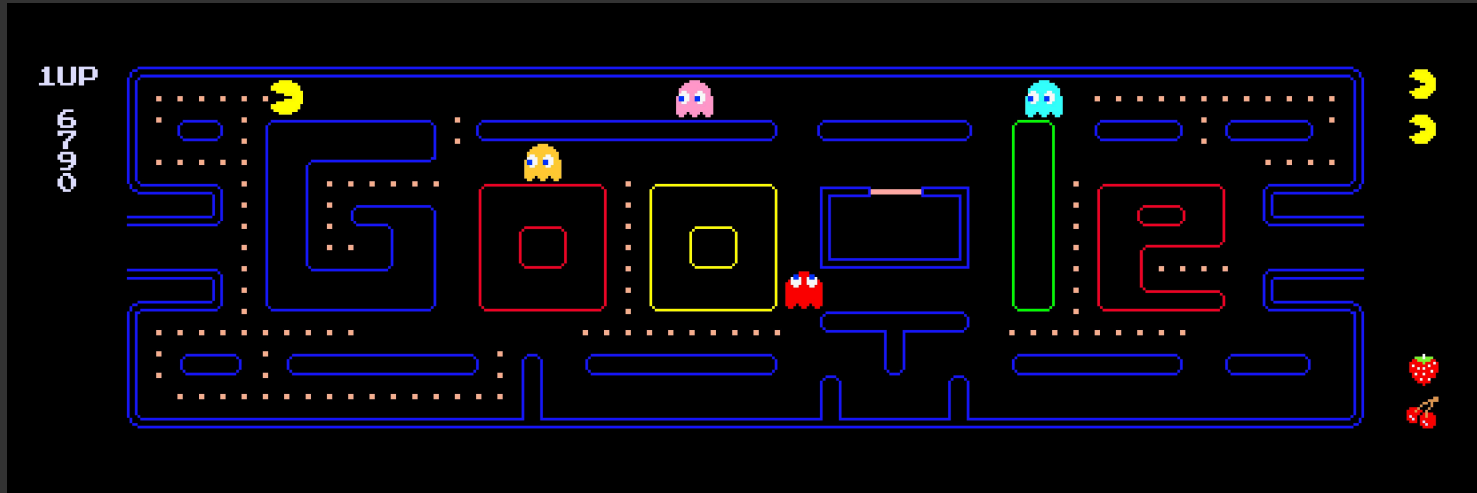
Search State: Keeps only details necessary for planning

Problem: Pathing

- States:
- Actions:
- Successor:
- Goal Test:

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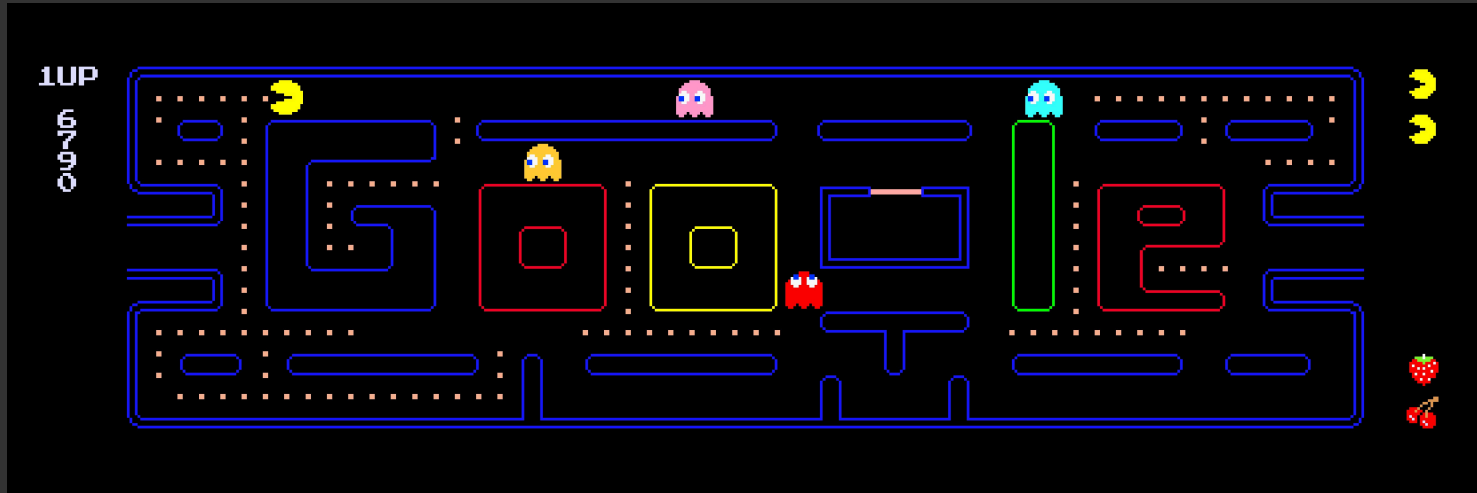
Search State: Keeps only details necessary for planning

Problem: Pathing

- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is $(x,y) == \text{END}$

WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



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Problem: Pathing

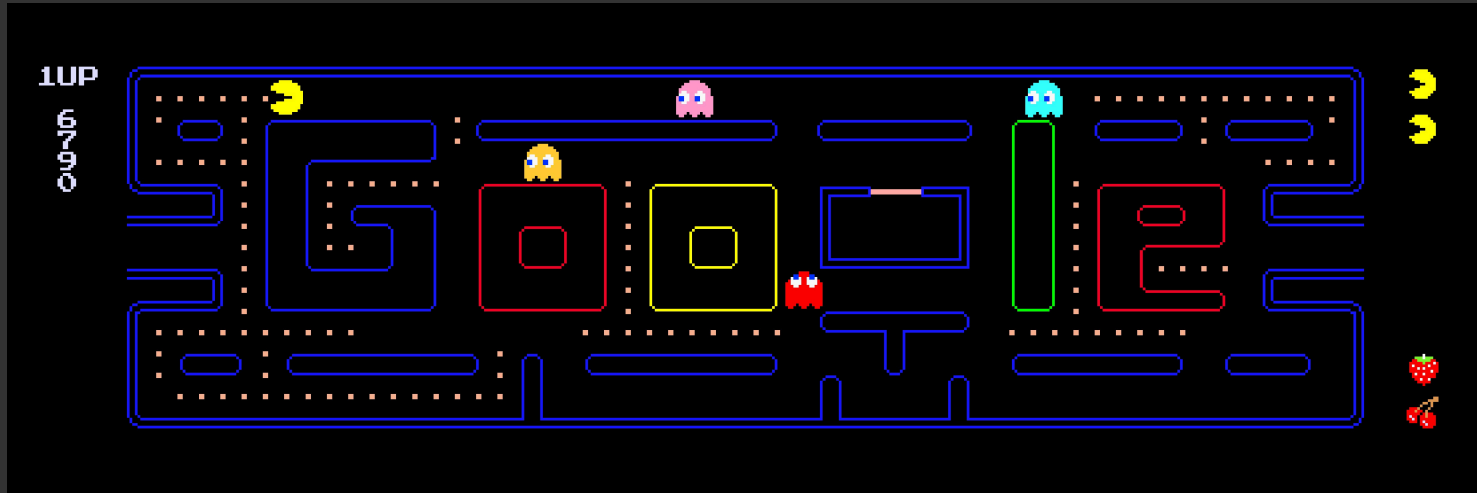
- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is $(x,y) == \text{END}$

Problem: Eat-All-Dots

- States:
- Actions:
- Successor:
- Goal Test:

WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



Search State: Keeps only details necessary for planning

Problem: Pathing

- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is $(x,y) == \text{END}$

Problem: Eat-All-Dots

- States: (x,y) , boolean for each dot
- Actions: NEWS
- Successor: update location, boolean for dots
- Goal Test: All dot booleans are false

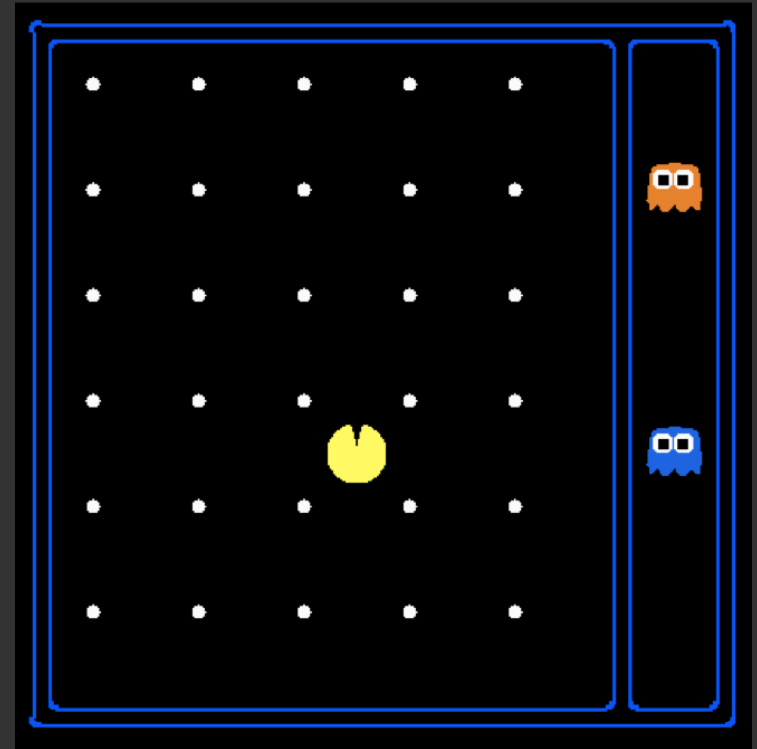
SEARCH SPACE SIZE

World State

- Agent Positions:
- Food Count:
- Ghost Positions:
- Agent Orientation:

Size

- World States:
- States for Pathing:
- States for Eat-All-Dots:



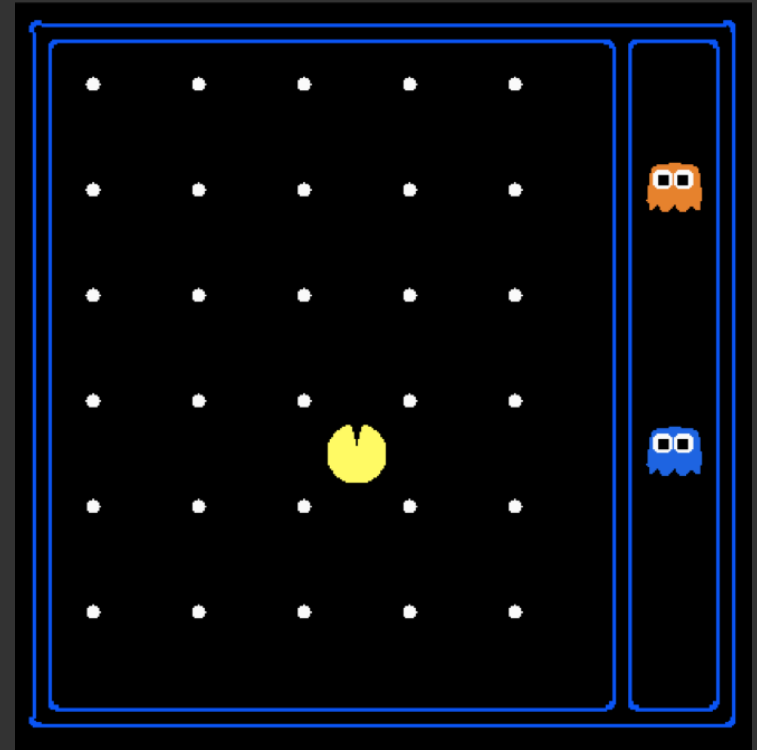
SEARCH SPACE SIZE

World State

- Agent Positions: 120
- Food Count: 30
- Ghost Positions: 12
- Agent Orientation: NEWS

Size

- World States: $120 \times 2^{30} \times 12^2 \times 4$
- States for Pathing: 120
- States for Eat-All-Dots: 120×2^{30}



Q & A



XKCD

